

Universität Freiburg – Institut für Physik

Applied Theoretical Physics

Prof. Dr. Michael Moseler

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Due: 22.06.2026 (Tuesday)

Computational Physics: Materials Science - Exercise 7, SS 2026

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Tutorials: Andreas Doell, Severin Sylla

The 2026 Sodium Cluster Challenge

Your main goal in this exercise will be to **win the 2026 sodium cluster challenge!** Generate the Na₇₆ cluster with the lowest potential energy among all students.

The potential energy has to be calculated using our Morse potential from exercise sheet 3 with the following parameters:

$$D = 0.06334 \text{ eV}, r_0 = 5.336 \text{ Å}, \alpha = 0.58993 \text{ Å}^{-1}, r_{\text{cut}} = 14.0 \text{ Å}. \quad (1)$$

Make sure that your cluster has enough vacuum around it such that it cannot interact with its periodic image.

Save your cluster as an ASE .traj file and hand it in together with your solution. Make sure the filename has the following form: Na76_cluster_<YOUR_NAME>.traj.

Feel free to try out different ways to minimize the energy. You should consider at least the following:

- (a) Set up a reasonable initial structure for your cluster. Run an MD simulation to heat your cluster and melt it.
- (b) To find a minimum potential energy configuration, try using some of the optimizers that are readily implemented in ASE (<https://ase-lib.org/ase/optimize.html>) to optimize the geometry of both your initial structure and the molten cluster. What are the differences between the optimizers? Examine the potential energy and the trajectories over the course of the optimization. Can you spot any differences between different optimization methods?
- (c) Can you tell whether your local minimum is somewhere close to a global minimum? What can you do to increase your confidence in this respect?
- (d) What could be another good strategy to find a low-lying minimum in potential energy using the tools we learned previously in this class? How would you do that in a real experiment? Try using temperature to your advantage.

What to submit

Hand in a zip file on ILIAS containing the following files:

- your Python source code
- a short report in PDF format with your main results
- your lowest energy Na₇₆ cluster in ASE .traj format

A good submission contains:

- a detailed description of your code and all the methods you have tried to tackle the sodium cluster challenge.
- plots that illustrate the dynamics of your MD simulations and how the energy is minimized.
- snapshots of your low-energy clusters using different methods

How to prepare for the exercise group

- Make sure you can explain your results and the code.
- Keep your chat history with an AI chatbot to explain in the tutorial how you **interacted** with it to develop your code.

Key takeaway questions

- What methods are used to locate local minima of the potential energy surface, and how do they differ from each other?
- How would you tackle the much harder task to search for a global minimum?
- What applications of geometry optimization do you know from the lecture? What useful information can be obtained from the optimized structures?